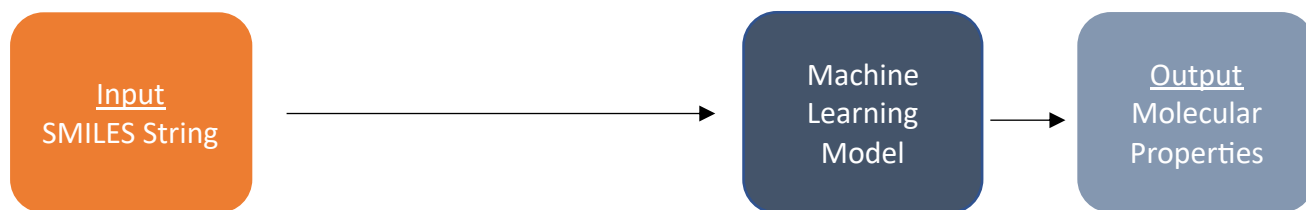
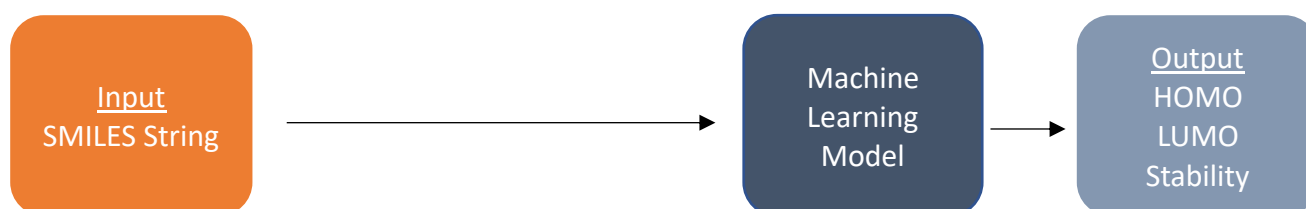


Task 1: Data Understanding

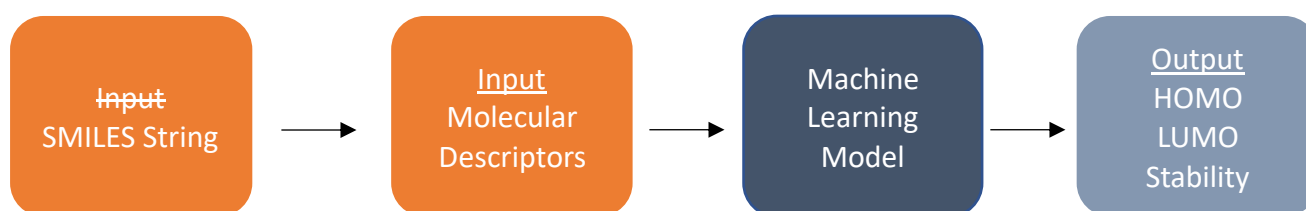
In the business understanding section, we learned that our goal is to predict molecular properties from SMILES strings.



The three properties we want to predict are HOMO, LUMO, and Stability. These are the outputs of our machine learning models. The inputs are the SMILES strings, which describe molecules in a text form.



However, **SMILES strings cannot be used directly as input in a machine learning model. They need to be transformed into numerical features first.** To do this, we use RDKit, a Python library that converts SMILES strings into molecular descriptors.



It is important to see the difference between **descriptors (input)** and **properties (output)**. A molecular descriptor is an input: a numerical or symbolic feature that encodes information about a molecule's structure. A molecular property is an output: a measurable characteristic that describes how the molecule behaves. Think of it like cars. Engine size, weight, and tire width are descriptors. Top speed and fuel efficiency are properties. In machine learning terms, descriptors are the features we feed into the model, while properties are the labels we try to predict.

In an ideal world, a dataset for this task would look like Table 1.

Table 1: The ideal format to train a machine learning model

SMILES	Molecular Descriptor 1	Molecular Descriptor 2	...	HOMO	LUMO	Stability

This would be perfect, but data science rarely starts this way. Instead, we often need to work with multiple raw datasets from different sources and shape them into the format we need by cleaning, transforming, and combining them.

Datasets

In this block, you will work with datasets from multiple sources.

Name	Molecule Count	Files	Contains
Starter	80	starter_dataset.xlsx	SMILES Molecular Descriptors Targets
HOPV	350	HOPV_homolumo.data HOPV_stability.csv	SMILES Targets
SQL	232	PostgreSQL DB	SMILES Targets
Kaggle	200	25261_classification_sample_submission.csv 25261_regression_sample_submission.csv	SMILES

There are a total of 350 molecules in HOPV, 232 in the SQL dataset, and 200 in Kaggle, which makes 782 molecules overall, of which **779 are unique molecules**. The starter dataset is a subset of the HOPV dataset. Now let's look at each dataset in more detail. For clarity, screenshots of the raw files are provided, with their filenames shown in the title bar at the top of each window.

To ensure a fair competition, the target values (HOMO, LUMO, and stability) for all datasets, including the Kaggle set, have been synthetically generated based on the real molecular properties. While they follow realistic chemical distributions, they will not match publicly available data for these specific molecules. Your goal is to model the relationships within the datasets we provide to you.

This dataset is modelling-ready and is prepared to help you begin quickly. It contains 80 molecules that are randomly selected from the 350 molecules in the HOPV dataset.

The first column of the file contains the SMILES string (Column A). The following columns are molecular descriptors (Columns B to HJ), [which can be found in RDKit](#). The final three columns are the target variables: HOMO, LUMO, and Stability (Columns HK to HM).

This dataset will only be used in Task 1. From Task 2 onward, you are expected to process the raw data yourself and bring it into the required format.

The Harvard Organic Photovoltaics (HOPV) Dataset

This dataset was created by researchers at Harvard University's Department of Chemistry and Chemical Biology. It contains SMILES strings as well as HOMO and LUMO values for 350 molecules. [Link to the paper.](#)

```
Cc1ccc(-c2c3cc(-c4ccc(-c5sc(-c6cccs6)c6cc(S(C)(=O)=O)sc65)s4)sc3c(-c3ccc(C)s3)c3ccsc32)s1
InChI=1S/C35H20Zn259/
c1-17-6-8-23(40-17)29-19-12-14-39-32(19)30(24-9-7-18(2)41-24)20-15-27(43-33(20)29)22-10-11-26(42-22)3
5-34-21(16-28(44-34)46(3,36)37)31(45-35)25-5-4-13-38-25/h-16H,1-3H3
10, 1021/ar5000743,BOUZZSMNJFUUCG-UHFFFAOYSA-
N, polymer, bulk, PC71BM, -5, 06, -3, 17, 1, 47, 1, 59, 7, 81, 0, 69, 17, 07, 66, 3
Cc1ccc(-c2c3cc(-c4ccc(-c5sc(-c6cccs6)c6cc(S(C)(=O)=O)sc65)s4)sc3c(-c3ccc(C)s3)c3ccsc32)s1
18
Conformer 1
68
C -2.8866535736 -6.5610552859 -0.6385425206
C -3.6289828835 -5.3595140545 -0.1290588703
C -4.5230238602 -5.2729895445 -0.9268907126
C -5.0512971275 -3.9614094769 1.1279707619
C -4.5599285317 -3.0187574579 0.2308948589
C -4.9132846545 -1.5953740514 0.1072940732
C -3.9354627018 -0.5623571511 -0.0356027933
C -2.5051872776 -0.6830085276 -0.0148121046
C -1.8343498457 -0.5253568888 -0.1288213292
C -0.4104169089 0.7690288489 -0.1400112537
C 0.2775743012 1.9852541003 -0.20700080564
C 1.6862461245 1.8482734380 -0.1886309381
C 2.1166009602 0.5165774679 -0.1068780206
C 3.4640994683 0.0134976216 -0.0625809377
S 3.8076161380 -1.6933272371 0.2229470232
C 5.5502213891 -1.5015062863 0.1135493056
C 6.4051666513 -2.6570261009 0.2500879232
C 6.0746072366 -4.0092033387 0.1067840765
C 7.1715075427 -4.8874453175 0.3366677619
C 8.3409124984 -4.2161719093 0.6530823025
```

The dataset `HOPV_homolumo.data` does not contain stability information. We created and provided this as a separate dataset, `HOPV_stability.csv`

[illegible]

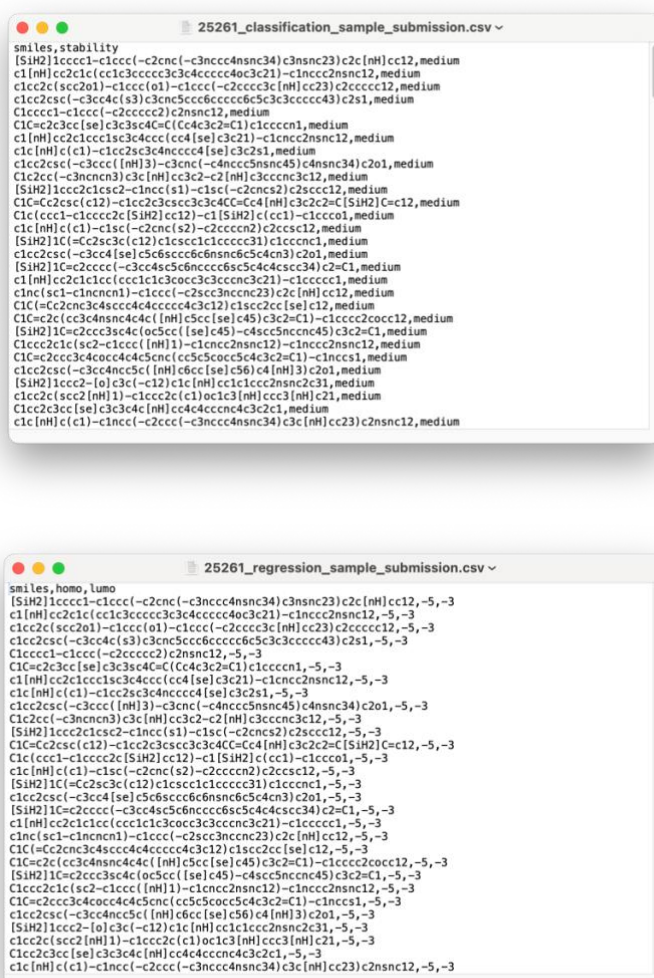
The SQL Dataset

A set of 232 molecules is provided in a PostgreSQL database as part of the SQL course. Instructions for connecting to and working with this database will be given in the corresponding task. You will first train your models using the HOPV dataset, and later incorporate this dataset to extend your experiments and evaluate its impact on model performance.

The Kaggle Dataset

The HOPV and Database molecules together form the development set for your classification and regression models. Once your models are ready, you will submit predictions for 200 SMILES molecules on Kaggle. The true values for these molecules are hidden, but after submission, you will see your results on the platform.

Two files are provided, one for the classification competition and one for the regression competition. These files contain 200 SMILES molecules each, with placeholder values in the target columns. Your task is to replace these placeholders with your model predictions and submit the files in the same format.



```
smiles, stability
[SIH2]1cccc1-clccc(-c2cnc(-c3nccc4nsnc34)c3nsnc23)c2c[nH]cc12, medium
c1[nH]cc2c1c(cc1c3cccc3c34cccc4oc3c21)-c1nccc2nsnc12, medium
c1cc2c(scc2o1)-clccc(o1)-clccc(-c2cccc3c[nH]cc23)c2cccc12, medium
c1cc2csc(-c3cc4c(s3)c3nc5ccc6cccc6c5c3c3cccc43)c2s1, medium
C1cccc1-clccc(-c2cccc2)c2nsnc12, medium
C1C=c2c3cc(se)c3c3sc4C=C(Cc4c3c2=C1)c1cccc1, medium
c1[nH]cc2c1ccc1sc3c4ccc(cc4[se]c3c21)-c1nccc2nsnc12, medium
c1c[nH]c(c1)-clcc2sc34ncccc4[se]c3c2s1, medium
c1cc2csc(-c3ccc([nH]3)-c3nc(-c4nccc5nsnc45)c4nsnc34)c2o1, medium
C1c2cc(-c3nccnc3)c3c[nH]cc3c2-c2[nH]c3ccnc3c12, medium
[SIH2]1ccc2c1csc2-clncc(s1)-c1sc(-c2cnsc2)c2sccc12, medium
C1C=Cc2csc(c12)-clcc2c3ccc3c3c4C=Cc4[nH]c3c2c2=C[SIH2]C=c12, medium
C1c(ccc1-clccc2c[SIH2]cc12)-c1[SIH2]c(cc1)-clccol, medium
c1c[nH]c(c1)-c1sc(-c2enc(s2)-c2cccc2)c2ccsc12, medium
[SIH2]1C(-C=C2sc3c(c12)c1sccl1cccc31)c1ccenc1, medium
c1cc2csc(-c3cc4[se]c5c6ccc6c6nsnc6c5c4c3c21)c2o1, medium
[SIH2]1C=c2cccc(-c3cc4sc5c6ncccc6c5c4c4sc34)c2=C1, medium
c1c[nH]cc2c1ccc(ccc1c1c3ccc3c3ccnc3c21)-c1cccc1, medium
c1nc(scl-clnccnc1)-clccc(-c2sc3nccnc23)c2c[nH]cc12, medium
C1C(-C=C2cnc34cccc4cccc4c3c12)c1sc2cc[se]c12, medium
C1C=c2c(cc3c4nsnc4c4c([nH]5cc[se]c45)c3c2=C1)-c1cccc2cccc12, medium
[SIH2]1C=c2ccc3sc4c(oc5cc([se]c45)-c4sc5nccnc45)c3c2=C1, medium
C1ccc2c1c(scc2-clccc([nH]1)-c1nccc2nsnc12)-c1nccc2nsnc12, medium
C1C=c2ccc3c4ccc4c4c5nc(c5c5ccc5c4c3c2c1)-c1nccs1, medium
c1cc2csc(-c3cc4ncc5c([nH]c6cc[se]c56)c4[nH]3)c2o1, medium
[SIH2]1ccc2-[o]c3c(-c12)c1c[nH]cc1c1ccc2nsnc2c31, medium
c1cc2c(scc2[nH]1)-c1ccc2c(c1)oc1c3[nH]ccc3[nH]c21, medium
C1cc2c3cc[se]c3c3c4c[nH]cc4c4ccnc4c3c2c1, medium
c1c[nH]c(c1)-c1ncc(-c2ccc(-c3nccc4nsnc34)c3c[nH]cc23)c2nsnc12, medium
```

```
smiles, homo, lumo
[SIH2]1cccc1-clccc(-c2cnc(-c3nccc4nsnc34)c3nsnc23)c2c[nH]cc12,-5,-3
c1[nH]cc2c1c(cc1c3cccc3c34cccc4oc3c21)-c1nccc2nsnc12,-5,-3
c1cc2c(scc2o1)-clccc(o1)-clccc(-c2cccc3c[nH]cc23)c2cccc12,-5,-3
c1cc2csc(-c3cc4c(s3)c3nc5ccc6cccc6c5c3c3cccc43)c2s1,-5,-3
C1cccc1-clccc(-c2cccc2)c2nsnc12,-5,-3
C1C=c2c3cc[se]c3c3sc4C=C(Cc4c3c2=C1)c1cccc1,-5,-3
c1[nH]cc2c1ccc1sc3c4ccc(cc4[se]c3c21)-c1nccc2nsnc12,-5,-3
c1c[nH]c(c1)-clcc2sc34ncccc4[se]c3c2s1,-5,-3
c1cc2csc(-c3ccc([nH]3)-c3nc(-c4nccc5nsnc45)c4nsnc34)c2o1,-5,-3
C1c2cc(-c3nccnc3)c3c[nH]cc3c2-c2[nH]c3ccnc3c12,-5,-3
[SIH2]1ccc2c1csc2-clncc(s1)-c1sc(-c2cnsc2)c2sccc12,-5,-3
C1C=Cc2csc(c12)-clcc2c3ccc3c3c4C=Cc4[nH]c3c2c2=C[SIH2]C=c12,-5,-3
C1c(ccc1-clccc2c[SIH2]cc12)-c1[SIH2]c(cc1)-clccol,-5,-3
c1c[nH]c(c1)-c1sc(-c2enc(s2)-c2cccc2)c2ccsc12,-5,-3
[SIH2]1C(-C=C2sc3c(c12)c1sccl1cccc31)c1ccenc1,-5,-3
c1cc2csc(-c3cc4[se]c5c6ccc6c6nsnc6c5c4c3c21)c2o1,-5,-3
[SIH2]1C=c2cccc(-c3cc4sc5c6ncccc6c5c4c4sc34)c2=C1,-5,-3
c1c[nH]cc2c1ccc(ccc1c1c3ccc3c3ccnc3c21)-c1cccc1,-5,-3
c1nc(scl-clnccnc1)-clccc(-c2sc3nccnc23)c2c[nH]cc12,-5,-3
C1C(-C=C2cnc34cccc4cccc4c3c12)c1sc2cc[se]c12,-5,-3
C1C=c2c(cc3c4nsnc4c4c([nH]5cc[se]c45)c3c2=C1)-c1cccc2cccc12,-5,-3
[SIH2]1C=c2ccc3sc4c(oc5cc([se]c45)-c4sc5nccnc45)c3c2=C1,-5,-3
C1ccc2c1c(scc2-clccc([nH]1)-c1nccc2nsnc12)-c1nccc2nsnc12,-5,-3
C1C=c2ccc3c4ccc4c4c5nc(c5c5ccc5c4c3c2c1)-c1nccs1,-5,-3
c1cc2csc(-c3cc4ncc5c([nH]c6cc[se]c56)c4[nH]3)c2o1,-5,-3
[SIH2]1ccc2-[o]c3c(-c12)c1c[nH]cc1c1ccc2nsnc2c31,-5,-3
c1cc2c(scc2[nH]1)-c1ccc2c(c1)oc1c3[nH]ccc3[nH]c21,-5,-3
C1cc2c3cc[se]c3c3c4c[nH]cc4c4ccnc4c3c2c1,-5,-3
c1c[nH]c(c1)-c1ncc(-c2ccc(-c3nccc4nsnc34)c3c[nH]cc23)c2nsnc12,-5,-3
```